

Tutorial problems for Sept 3

- 1) page 79: Show that $\{(x_1, x_2, x_3) \in \mathbb{R}^3 / x_1 + x_2 + x_3 > 0\}$ is open
- 2) page 81 (triangle problem)
- 3) page 83
- 4) page 88
- 5) Sketch the following loci in $\mathbb{R}^3$

$$\left. \begin{aligned} z &= x^2 + y^2 \\ z &= x^2 - y^2 \\ z^2 &= x^2 + y^2 \\ z^2 &= x^2 + y^2 + \alpha^2 \\ z^2 &= x^2 + y^2 - \alpha^2 \end{aligned} \right\} \alpha \neq 0.$$

- 6) Sketch the following loci in $\mathbb{R}^2$

$$\frac{y^2}{2} - \frac{g}{l} \sin x = E \quad (E \text{ is a const)}$$

$$((x-a)^2 + y^2)((x+a)^2 + y^2) = \text{const}.$$

Find points of Max/Min/Saddle.
 (Not covered in lecture yet but a

Sneak preview would be good

You'll notice that when the constant c goes ~~thru~~ past a critical value the level set $f(x, y) = c$ undergoes a dramatic change